

Multi-Cohort Simulation Study: Shared vs. Study-Specific Factors (Proposal)

Andrew Walther

2026-06-14

1. Purpose and Background

Prior synthetic validation of the Supervised Bayesian Matrix Factorization (SSBMF) model was carried out only in a **single-cohort** setting. The real application pools several PDAC cohorts that differ both by **platform** (RNA-seq, microarray, proteomics) and by **biology**. In that setting the latent factors fall into two kinds:

- **Shared factors** — gene-expression programs present across *all* studies. These are the cross-cohort biological signal we want to detect and, where they are prognostic, to use for survival prediction.
- **Study-specific factors** — programs private to a *single* study. These absorb platform/batch effects and cohort-private biology that should *not* be mistaken for shared, generalizable signal.

This proposal specifies a simulation that generates data with a **known** shared / study-specific factor structure, so we can test whether the current model **natively** recovers that distinction — the same goal the previously discussed cohort dummy-indicator approach was meant to serve. If the base model already separates shared from study-specific factors, the dummy indicator columns may be unnecessary (meeting notes, May 29 2026).

The design follows three decisions from the May 29 meeting:

1. **EBMF-grounded ground truth.** Realistic gene programs are taken from an Empirical Bayes Matrix Factorization (flashier) fit to the merged real training cohorts. EBMF is also our **unsupervised benchmark** (Section 6).
2. **Shared factors carry survival signal; study-specific factors do not** ($\beta \neq 0$ on shared, $\beta = 0$ on specific). Survival is generated from the shared component only.
3. **The signal-strength of shared vs. study-specific structure is tunable** through the magnitude of the F columns, which defines the three scenarios.

2. Notation

Symbol	Meaning	Shape
C	number of cohorts (studies)	scalar
n_c	patients in cohort c ; $n = \sum_c n_c$	scalar
p	number of genes	scalar
K	total number of latent factors	scalar
$\mathcal{S}$	index set of shared factors	$ \mathcal{S} $
$\mathcal{P}_c$	index set of factors specific to cohort c	$ \mathcal{P}_c $
L	patient loadings	$n \times K$
F	gene programs (factors)	$p \times K$
β	log-hazard coefficients	K
Y	observed genomics matrix	$n \times p$

Rows of Y (and L) are **stacked by cohort**: rows $1..n_1$ are cohort 1, the next n_2 rows are cohort 2, and so on. Factors partition disjointly: $K = |\mathcal{S}| + \sum_c |\mathcal{P}_c|$.

3. Data-Generating Process

3.1 Loadings L ($n \times K$) — block structure encodes specificity

Every loading is drawn non-negative,

$$L_{ik} \sim \text{Exponential}(\text{rate} = 1), \quad \mathbb{E}[L] = 1, \quad \mathbb{E}[L^2] = 2,$$

matching the point_exponential prior used by the fitters. Specificity is then imposed by **zeroing out-of-cohort loadings**:

- **Shared factor** $k \in \mathcal{S}$: keep L_{ik} for all patients i .
- **Cohort- c -specific factor** $k \in \mathcal{P}_c$: set

$$L_{ik} = 0 \quad \text{for every } i \notin \text{cohort } c.$$

A study-specific factor therefore contributes to Y **only in its own cohort's rows** — this is the operational definition of “study-specific.” The resulting L is block-sparse: shared columns are dense, specific columns are zero outside one cohort block.

3.2 Gene programs F ($p \times K$) — realism + the signal knob

Each column $F_{\cdot k}$ is a **gene program**, normalized to unit L_2 norm so its scale is set explicitly by an amplitude (below):

- **Preferred (EBMF-grounded)**: a column of `ldf(flash_fit, type=2)$F` fit to the merged real training cohorts (realistic sparse programs), made **non-negative** (absolute value) to match the model's non-negative point_exponential factor prior. When K exceeds the EBMF rank, columns are recycled by index. *Matching the prior to the data sign structure is essential for a fair test*: signed programs against a non-negative prior artificially depress recovery. The EBMF benchmark is likewise run with a non-negative prior.
- **Synthetic fallback** (no real data / no flashier): for each factor pick $\lceil \text{active_rate} \cdot p \rceil$ active genes (default 5%) and set their weight to $1/\sqrt{\#\text{active}}$ (unit-norm column).

Signal-strength knob. Multiply each column by an amplitude a_k : a_{shared} for shared columns, a_{specific} for specific ones. Since the variance factor k contributes to Y is $\mathbb{E}[L_{\cdot k}^2] \|F_{\cdot k}\|^2 a_k^2$ and $\|F_{\cdot k}\|^2 = 1$, the **variance ratio scales as** $(a_{\text{shared}}/a_{\text{specific}})^2$. We call $\text{shared_signal_ratio} = a_{\text{shared}}/a_{\text{specific}}$ the continuous dial behind the scenarios in Section 4.

3.3 Genomics model (Gaussian)

$$Y = LF^\top + \text{offset} + E, \quad E_{ij} \sim \mathcal{N}(0, 1/\tau_j), \quad \tau_j \sim \text{Gamma}(\text{shape} = 2, \text{rate} = 2).$$

- Per-gene precision τ_j gives gene j noise SD $1/\sqrt{\tau_j}$ ($\mathbb{E}[\tau_j] = 1$), matching the established single-cohort DGP.
- **offset** (default **0**): per the meeting notes, batch effects are absorbed by the study-specific component by construction, so an explicit offset is usually unnecessary. When enabled, one vector per cohort $o_c \sim \mathcal{N}(0, \sigma_{\text{offset}}^2 I_p)$ is added to that cohort's rows (a pure platform shift) for stress-testing.
- Y is **column-centered** before fitting (standard preprocessing).

3.4 Survival model (Weibull proportional hazards) — shared factors only

$$\eta_i = \sum_{k \in \mathcal{S}} L_{ik} \beta_k \quad (\beta_k = 0 \text{ for every specific factor}),$$

$$T_i = \left(\frac{-\log U_i}{\text{scale0} \cdot \exp(\eta_i)} \right)^{1/\text{shape}}, \quad U_i \sim \text{Unif}(0, 1), \quad \text{shape} = 1.5, \quad \text{scale0} = 0.01.$$

The baseline cumulative hazard is Weibull, $H_0(t) = (t \cdot \text{scale0})^{\text{shape}}$. The default shared coefficients cycle the established pattern $(1.5, -1.2, 0.8, -0.5)$ to length $|\mathcal{S}|$. **Censoring** is calibrated to $\approx 30\%$ by bisection: $\text{censor_base} \sim \text{Exp}(1)$, $\text{time} = \min(\text{event}, \text{censor})$, $\text{status} = \mathbf{1}\{\text{event} \leq \text{censor}\}$.

Anchoring caveat (meeting notes). Because the survival outcome is linked only to shared factors, the supervision “anchors” those factors into the shared component. This is exactly the intended mechanism; the results report discusses whether it produces cleaner shared/specific separation.

4. The Three Scenarios

The scenarios differ only in how the K factors are partitioned into shared vs. specific (and, optionally, in `shared_signal_ratio`). For the primary setting $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$:

Scenario	$ \mathcal{S} $ shared ($\beta \neq 0$)	specific per cohort ($\beta = 0$)	Interpretation
All shared	6	0, 0	Studies effectively one cohort
Hybrid	2	2, 2	Both shared and private structure
Nothing shared	0	3, 3	No cross-cohort structure

Nothing-shared is a deliberate false-positive control. With $\mathcal{S} = \emptyset$, $\eta \equiv 0$ and survival carries no factor signal. The correct result is C-index ≈ 0.5 and all $|\hat{\beta}|$ below threshold — the model should *not* fabricate a shared prognostic factor. An optional knob (`specific_prognostic`, default off) can give each cohort a within-cohort prognostic factor if we later want survival signal in this regime.

Worked example — Hybrid, $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$

- **Dimensions:** $n = 300$; Y is 300×1000 , L is 300×6 , F is 1000×6 , β is length 6. Rows 1–150 = cohort 1; 151–300 = cohort 2.
- **Factor labels:** [shared, shared, specific_1, specific_1, specific_2, specific_2].
- **Loadings L :** columns 1–2 dense (all 300 patients). Columns 3–4 zero for rows 151–300 (private to cohort 1). Columns 5–6 zero for rows 1–150 (private to cohort 2). L shows a visible block-sparse pattern.
- **Coefficients:** $\beta = [1.5, -1.2, 0, 0, 0, 0]$ — only the two shared factors drive survival.
- **What a correct model should find:** factors 1–2 with loadings across *both* cohorts *and* nonzero $\hat{\beta}$; factors 3–6 cohort-private with $\hat{\beta} \approx 0$. The unsupervised EBMF benchmark should recover the same six gene programs but cannot say which two are prognostic.

Secondary sensitivity: a 3-cohort hybrid (2 shared + 1 private per cohort) and a `shared_signal_ratio` sweep within the hybrid regime.

5. Integration with the SSBMF (YFB) Model

The DGP emits exactly the inputs the fitters expect — Y ($n \times p$, column-centered), `time`, `status`, and `cohort_id` (a factor). These map directly onto:

- `fit_cox_on_yf()` — the **YFB** model, linear predictor $\eta = (YF)\beta$. This is the recommended model; `EF_norms` from the fit must be passed to `predict_cox_on_yf()` for consistent train/test scaling.
- `fit_supervised_mf_modular()` — the **LB** model, $\eta = L\beta$ (retained for comparison).

Best-practice configuration (D4-aligned): `prior_beta = "normal"` (avoids the $\beta \rightarrow 0$ collapse), `alpha = 0.5`, `lambda = 1.0`. Per-platform normalization — the single most important preprocessing step on real merged data — is implicitly satisfied here because every cohort block is generated on a common scale; a per-cohort-centering (“z-score”) arm is retained as a reference. The cohort dummy indicator (`cohort_id`) uses **corner-point encoding** ($C - 1$ fixed binary columns that absorb per-cohort gene means and never enter the Cox predictor).

The native-identification test. The headline analysis fits **without** the dummy indicator (`cohort_id = NULL`) and asks whether the fitted factors separate into shared vs. study-specific on their own. The $\pm$ dummy comparison then quantifies whether the indicator adds anything.

6. Supervised vs. Unsupervised (EBMF) Benchmark

EBMF (flashier, greedy + backfit) is fit to the **same simulated Y** with **no survival information**, and scored on the *identical* factor-recovery and specificity-classification metrics as the supervised fits. Because EBMF is survival-blind, it has **no β** . The comparison answers two questions:

1. Does the supervised model recover the shared/study-specific structure *as well as* the unsupervised gold standard (i.e., supervision does not hurt the factorization)?
2. Does survival supervision *additionally* yield correct β — flagging which recovered factors are prognostic — which EBMF fundamentally cannot?

This is the value-add argument for the paper: same structural recovery, plus correct survival attribution.

Note: two distinct EBMF uses. The **template-building** EBMF (Section 3.2) is fit to *real* data to manufacture realistic ground-truth gene programs. The **benchmark** EBMF here is fit to each *simulated Y* and scored as a competing method. They do not share a fit.

7. Evaluation Metrics

For each (scenario $\times$ model arm $\times$ seed) we record:

- **Shared-factor recovery** — mean best $|\text{cor}|$ between estimated and true gene programs (F columns, gene space, comparable across LB/YFB/EBMF), over the *true shared* factors.
- **Specific-factor recovery** — same, over the *true study-specific* factors.
- **Specificity-classification accuracy** — label each *estimated* factor shared/specific from its per-cohort loading energy, compare to the truth. This is the direct test of “does the model natively identify the distinction?”
- **Survival-effect recovery** — $|\hat{\beta}|$ on factors matched to true shared (expect $\$ > \$$ threshold) vs. true specific (expect ≈ 0). Reported as true-positive and **false-positive prognostic rates** (the latter is the anchoring / nothing-shared check). Threshold from `globals.yml`: `k_selection$beta_threshold`.
- **C-index** — orientation-free held-out concordance, $\max(C, 1 - C)$.

Comparisons reported: (i) YFB vs. LB; (ii) base vs. +cohort-dummy (Δ in each metric — “are dummies necessary?”); (iii) supervised vs. unsupervised EBMF.

8. Reference Data-Generating Code

The following is the complete, self-contained DGP (the project implementation wraps this same logic with configuration and seeds). A reader can run it directly given the small `.calibrate` helper.

```
# Bisection calibration of censoring scale to a target censoring fraction.
.calibrate <- function(ev, base, target = 0.30, n_iter = 40) {
  lo <- 1e-3; hi <- max(ev) * 10
  for (i in seq_len(n_iter)) {
    mid <- sqrt(lo * hi)
    if (mean(base * mid < ev) > target) lo <- mid else hi <- mid
  }
  hi
}

generate_multicohort_data <- function(
  C          = 2L,          # number of cohorts
  n_per      = rep(150L, C), # patients per cohort (rows stacked by cohort)
  p          = 1000L,       # genes
  K_shared   = 2L,          # # shared factors      (beta != 0)
```

```

K_specific = rep(2L, C),      # # specific factors per cohort (beta = 0)
F_templates = NULL,          # p x K_ebmf EBMF programs; NULL -> synthetic sparse
a_shared = 1.0,              # amplitude of shared F columns
a_specific = 1.0,            # amplitude of specific F columns
active_rate = 0.05,          # synthetic-fallback sparsity
beta_shared = NULL,          # length K_shared; NULL -> recycle c(1.5,-1.2,0.8,-0.5)
offset_sd = 0.0,             # explicit platform offset SD (0 = off)
shape = 1.5, scale0 = 0.01,  # Weibull-PH baseline
target_cens = 0.30,          # target censoring fraction
seed = 1L) {

set.seed(seed)
n <- sum(n_per)
K <- K_shared + sum(K_specific)
cohort_id <- factor(rep(seq_len(C), times = n_per))

# label + owner per column (owner 0 = shared)
labels <- c(rep("shared", K_shared),
            unlist(lapply(seq_len(C), function(cc) rep(paste0("specific_", cc), K_specific[cc]))))
owner <- c(rep(0L, K_shared),
           unlist(lapply(seq_len(C), function(cc) rep(cc, K_specific[cc]))))

# L (n x K): Exponential(1), block-zeroed outside owner cohort
L <- matrix(rexp(n * K, rate = 1), n, K)
for (k in seq_len(K)) if (owner[k] != 0L) L[cohort_id != owner[k], k] <- 0

# F (p x K): EBMF templates (unit-norm) or synthetic sparse (unit-norm)
Fm <- matrix(0, p, K)
if (!is.null(F_templates)) {
  idx <- ((seq_len(K) - 1L) %>% ncol(F_templates)) + 1L
  Fm <- F_templates[, idx, drop = FALSE]
  Fm <- sweep(Fm, 2, sqrt(colSums(Fm^2)) + 1e-10, "/")
} else {
  for (k in seq_len(K)) {
    active <- sample.int(p, max(1L, round(active_rate * p)))
    Fm[active, k] <- 1 / sqrt(length(active))
  }
}
amp <- ifelse(labels == "shared", a_shared, a_specific) # signal-strength knob
Fm <- sweep(Fm, 2, amp, "*")

# per-gene Gaussian noise: tau_j ~ Gamma(2,2)
tau <- rgamma(p, shape = 2, rate = 2)
E <- sweep(matrix(rnorm(n * p), n, p), 2, sqrt(tau), "/")

# optional explicit platform offset (off by default)
offset <- matrix(0, n, p)
if (offset_sd > 0) for (cc in seq_len(C)) {
  ov <- rnorm(p, 0, offset_sd)
  offset[cohort_id == cc, ] <- matrix(ov, sum(cohort_id == cc), p, byrow = TRUE)
}

# assemble + column-center
Y <- L %*% t(Fm) + offset + E
Y <- sweep(Y, 2, colMeans(Y), "-")

# survival from SHARED factors only
if (is.null(beta_shared))

```

```

    beta_shared <- if (K_shared > 0) rep_len(c(1.5, -1.2, 0.8, -0.5), K_shared) else numeric(0)
    beta <- numeric(K); beta[labels == "shared"] <- beta_shared
    eta <- as.vector(L %*% beta) # == 0 in the "nothing shared" scenario

    event <- (-log(runif(n)) / (scale0 * exp(eta)))^(1 / shape)
    cb <- rexp(n, rate = 1)
    cs <- .calibrate(event, cb, target = target_cens)
    ctime <- cb * cs
    time <- pmin(event, ctime)
    status <- as.integer(event <= ctime)

    list(Y = Y, time = time, status = status, cohort_id = cohort_id,
         L_true = L, F_true = Fm, beta_true = beta,
         factor_labels = labels, factor_owner = owner)
}

```

9. Open Questions for Discussion (Rashid / Yusha)

1. **Specificity definition.** We encode study-specific factors via block-zero loadings (a factor is active in exactly one cohort). Is this the structure you intend, or should specific factors also have *non-overlapping gene programs* per cohort?
2. **Nothing-shared survival.** Default leaves survival null in the nothing-shared scenario (false-positive control). Do you prefer instead a within-cohort prognostic specific factor so each cohort has its own survival signal?
3. **Cohort count.** Primary is $C = 2$ (matches the real training data and the fixed-effects rationale); a $C = 3$ hybrid is included as sensitivity. Is 2 sufficient for the methods claim, or should 3 be primary?
4. **Signal ratio.** Default $a_{\text{shared}} = a_{\text{specific}}$. Should the hybrid scenario stress a regime where study-specific variance *dominates* shared variance (the harder, more realistic case)?

This is the design to be reviewed before the implementation results are finalized. The accompanying results report (working draft) demonstrates the pipeline on this design and will be revised per feedback.